USER INTERFACE DESIGN (CS23A34)

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EXP NO:4 B) Task Analysis for an APP using DIA

AIM :

The aim of this experiment is to perform **task analysis** for an application (online shopping app), identify user interactions, document **user flows**, and create corresponding **wireframe diagrams** using Dia software.

TOOL USED :

- **Dia Diagram Editor**
- Official Website: http://dia-installer.de/
- Purpose: Used to create flowcharts, wireframes, and system diagrams

THEORY :

Task analysis is the process of understanding how users perform tasks within a system. It helps in identifying:

- User goals
- Steps involved in completing a task
- Decision points and actions

User flow diagrams visually represent the sequence of steps a user follows to complete a task. Wireframes are simple visual layouts of application screens that show structure and navigation without detailed design.

In this experiment, an **online shopping app** is used to analyze how a user browses products, adds items to the cart, and completes a purchase.

PROCEDURE :

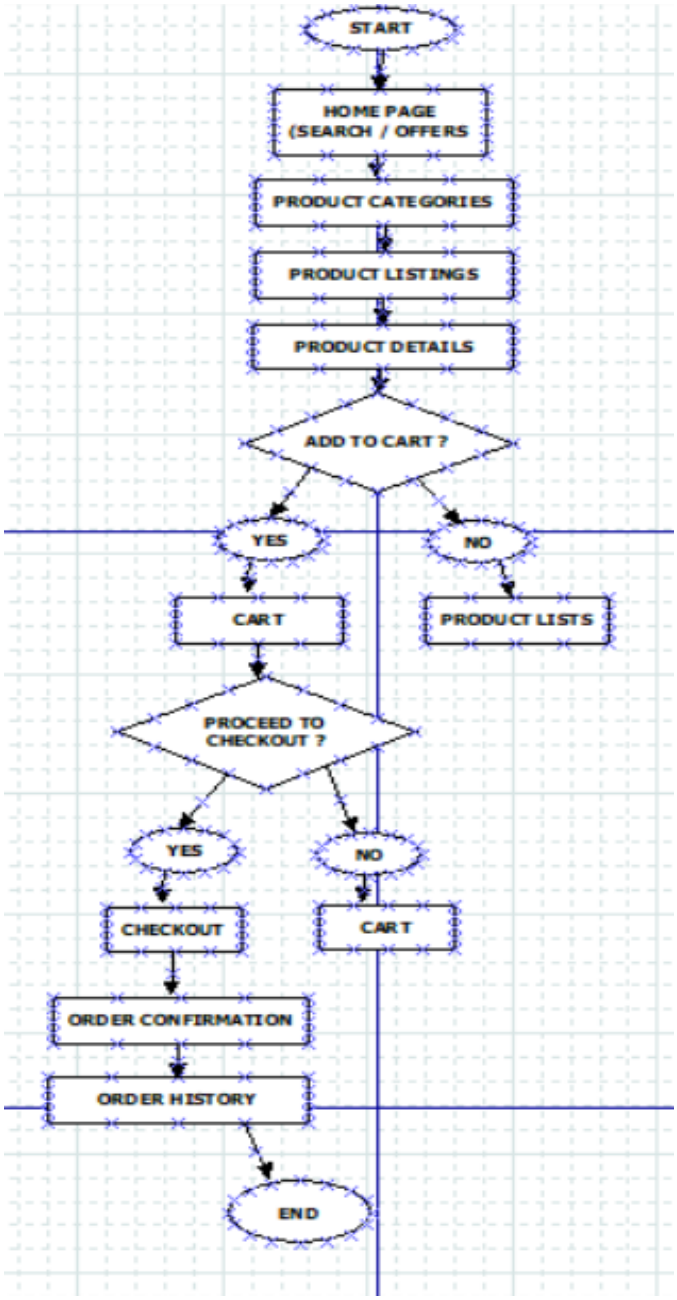
1. **Install Dia**

- Download Dia from the official website
- Install and launch the application

2. **Create New Diagram**
    - Go to *File → New*
    - Select *Flowchart*
3. **Add Shapes**
    - Use:
        - Oval → Start/End
        - Rectangle → Screens (Home, Cart, etc.)
        - Diamond → Decision points
4. **Design User Flow**
    - Create flow for:
        - Home Page
        - Product Categories
        - Product Listings
        - Product Details
        - Cart
        - Checkout
        - Order Confirmation
        - Order History
5. **Connect Shapes**
    - Use arrows to connect all steps
    - Add labels like “Yes” / “No” for decisions

USER FLOW DIAGRAM :

START → Home Page → Product Categories → Product Listings → Product Details → Decision (Add to Cart?) → Cart → Decision (Proceed to Checkout?) → Checkout → Order Confirmation → Order History → END



INDIVIDUAL USER FLOW DIAGRAMS :

1. Browsing Flow

Home Page → Categories → Product Listings

2. Product Selection Flow

Product Listings → Product Details

3. Purchase Flow

Product Details → Cart → Checkout → Order Confirmation

4. Order Tracking Flow

Order Confirmation → Order History

WIREFRAMES :

Wireframes represent the basic layout of each screen:

- Home Page → Search bar, categories
- Categories Page → List of product types
- Product Listings → Product grid/list
- Product Details → Image, price, description
- Cart → Selected items
- Checkout → Address & payment
- Order Confirmation → Success message
- Order History → List of past orders

RESULT :

The task analysis and user flow for an online shopping application were successfully created. The user journey was clearly visualized using flowcharts and wireframes in Dia, helping to understand the interaction between different components of the system.

CONCLUSION :

This experiment helped in understanding how users interact with an application and how task analysis improves usability. Creating user flows and wireframes provides a clear structure for designing user-friendly interfaces.